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B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2024
Third Semester

21CSC202J – OPERATING SYSTEMS

(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Answer **ALL** Questions

Marks BL CO PO

- | | | | | |
|--|---|---|---|----|
| 1. _____ of the following is NOT an operating system? | 1 | 1 | 1 | 1 |
| (A) Oracle | | | | |
| (B) Windows | | | | |
| (C) Linux | | | | |
| (D) DOS | | | | |
| 2. BIOS is used by _____. | 1 | 1 | 1 | 2 |
| (A) Compiler | | | | |
| (B) Operating system | | | | |
| (C) Application software | | | | |
| (D) Interpreter | | | | |
| 3. When several processes access the same data concurrently and the outcomes of the execution depends on the particular order in which the access takes place is called _____. | 1 | 1 | 1 | 1 |
| (A) Essential condition | | | | |
| (B) Critical condition | | | | |
| (C) Race condition | | | | |
| (D) Dynamic condition | | | | |
| 4. Identify the process that can be affected by other process executing in the system. | 1 | 1 | 1 | 2 |
| (A) Child process | | | | |
| (B) Parent process | | | | |
| (C) Init process | | | | |
| (D) Cooperating process | | | | |
| 5. Find the major problem with priority scheduling. | 1 | 1 | 2 | 1 |
| (A) Starvation | | | | |
| (B) Low priority | | | | |
| (C) Deadlock | | | | |
| (D) Definite blocking | | | | |
| 6. Performance of the CPU is measured by _____. | 1 | 1 | 2 | 2 |
| (A) Flaps | | | | |
| (B) GHz | | | | |
| (C) Throughput | | | | |
| (D) MHz | | | | |
| 7. _____ is a solution to the problem of low-priority processes being blocked indefinitely. | 1 | 1 | 2 | 11 |
| (A) Aging | | | | |
| (B) Starvation | | | | |
| (C) Deadlocks | | | | |
| (D) Wait queue | | | | |
| 8. A _____ stores the state of a process, so that it can be recovered and resumed later. | 1 | 1 | 2 | 1 |
| (A) Multi programming | | | | |
| (B) Context switch | | | | |
| (C) State | | | | |
| (D) Uni programming | | | | |

9. The purpose of the disks swap space is to _____.
 (A) Saving temporary HTML pages (B) Storing the super block
 (C) Storing device drivers (D) Saving process data
10. Page fault occurs _____.
 (A) When a requested page is in memory (B) When a page is corrupted
 (C) When a requested page is not in memory (D) When an exception is thrown
11. _____ performs run time mapping from virtual to physical addresses.
 (A) CPU (B) PCI
 (C) PCB (D) Memory Management Unit
12. The _____ points to the address of a page table in memory.
 (A) Page table base register (B) Stack pointer
 (C) Program counter (D) Page register
13. _____ is a memory management strategy that involves the system storing and retrieving data from secondary storage for usage in main memory.
 (A) Mapping (B) Fragmentation
 (C) Paging (D) Buffer
14. A solution to the problem of external fragmentation is _____.
 (A) Larger memory space (B) Smaller memory space
 (C) Compaction (D) Secondary memory space
15. The page table contains
 (A) Page offset (B) Page size
 (C) Logical addresses (D) Base address of each page in physical memory
16. _____ of the following statements deals with associative memory.
 (A) There is no need for an address (B) The address of the data is supplied by the users
 (C) The data are accessed sequentially (D) The address of the data is generated by the CPU
17. Information about a file can be found in the _____ include the owner, permission in location of the file.
 (A) Device drivers (B) File
 (C) File control block (FCB) (D) File system
18. _____ is the file system used by windows 95.
 (A) FAT 32 (B) FAT 16
 (C) LMFS (D) NTFS
19. For real-time operating systems, interrupt latency should be _____.
 (A) Zero (B) Minimal
 (C) Dependent on the scheduling (D) Maximum
20. _____ is one that processes data instructions without any delay.
 (A) Real-time system (B) Instruction system
 (C) Offline system (D) Online system

PART – B (5 × 8 = 40 Marks)Answer **ALL** Questions

Marks BL CO PO

21. a. Describe the different services provided by the operating system and their significance in managing computer resources effectively. 8 3 1 3

(OR)

- b. Recall the concept of system calls and elaborate different types of system calls that various applications can make to request services from the operating system. 8 2 1 5
22. a. Illustrate the three types of thread models using diagrams and explain how they influence program execution and resource management in modern operating systems. 8 3 2 5

(OR)

- b. Demonstrate how synchronization hardware handles critical section problems, ensuring mutual exclusion and preventing race conditions in concurrent processes. 8 3 2 3
23. a. Explain the First Come First Served (FCFS) and Shortest Job First (SJF) disk scheduling algorithms with examples and highlight their advantages and limitations. 8 3 3 4

(OR)

- b. Describe how the wait-for graph is utilized in detecting and resolving deadlocks in concurrent systems with an example. 8 3 3 5
24. a. Discuss the concept of contiguous memory allocation in operating system. Explain its disadvantages. 8 2 4 5

(OR)

- b. Elaborate the linked allocation and indexed allocation methods used in file systems, outlining their structures and performance characteristics. 8 3 4 5
25. a. Discuss how the access matrix is utilized in enforcing protection policies and controlling access to system resources in multi-user environments. 8 2 5 4

(OR)

- b. Consider the statement “A capability-based system provides greater flexibility than a ping protection scheme in enforcing protection policies”. Justify. 8 3 5 3

PART – C (1 × 15 = 15 Marks)Answer **ANY ONE** Question

Marks BL CO PO

26. Consider the following processes with its arrival time, burst time and priority. 15 2 3 4

Process	Arrival time	Burst time	Priority
P1	0	12	2
P2	1	16	5
P3	2	5	1
P4	3	8	4
P5	4	12	3

Sketch Gantt chart and compute average turn around time and average waiting time for the following scheduling algorithm:

- (i) First Come First Served (FCFS)
- (ii) SJF (Shortest Job First)
- (iii) Preemptive Scheduling
- (iv) Round Robin (RR)

27. Consider the following page-reference string:

15 3 4 3

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2

Compute how many page faults would occur for the following Page Replacement algorithms. The total frame size is 4 and frames are initially empty.

- (i) FIFO page replacement
- (ii) Optimal page replacement
- (iii) LRU page replacement

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